

Compute $\sum_{n=1}^{\infty} \frac{n^5}{e^{2\pi n} - 1}$

Solution: Let $f(z) = \frac{z^5}{(e^{2\pi z} - 1)(e^{-2\pi iz} - 1)}$ and $g(z) = \frac{z^5}{e^{2\pi z} - 1}$, then $f(z) - if(iz) = g(z)$, and

$$\int_0^{\infty} g(x) dx = \frac{1}{(2\pi)^6} \int_0^{\infty} \frac{t^{6-1}}{e^t - 1} \frac{dt}{t} = \frac{1}{(2\pi)^6} \Gamma(6) \zeta(6) = \frac{1}{504}.$$

(Recall $\Gamma(s) \zeta(s) = \int_0^{\infty} \frac{t^{s-1}}{e^t - 1} dt$).

Note that $f(z)$ has simple poles at $z = n, in$ where $n \in \mathbb{Z} \setminus \{0\}$, and

$$\text{Res}(f, n) = \text{Res}(f, in) = -\frac{1}{2\pi i} \frac{n^5}{e^{2\pi n} - 1}.$$

Let $C_{\varepsilon}^n = \{z : |z - n| = \varepsilon\}$ for any n, ε , and for $R > 0$ ($R \in \mathbb{Z} + 1/2$), let $\gamma_1 = \{\text{Re}^{i\theta} : 0 \leq \theta \leq \frac{\pi}{4}\}$, γ_2 be the interval $[0, R]$ with semicircles of radii ε at each $1 \leq n < R$, γ_3 be the line $[0, iR]$ with semicircles of radii ε at each $in, 1 \leq n < R$. Since f has no poles in this region,

$$0 = \int_{\gamma_1} f dz + \int_{\gamma_2} f dz - \int_{\gamma_3} f dz. \quad (1)$$

Clearly $\int_{\gamma_1} f dz \rightarrow 0$ as $R \rightarrow \infty$, since the denominator has exponential growth. Let γ_4 be the reflection of γ_2 through the real line, then

$$\int_{\gamma_4} f dz - \int_{\gamma_3} f dz = \int_{\gamma_4} f(z) - if(iz) dz = \int_{\gamma_4} g(z) dz = \int_0^{\infty} g(x) dx. \quad (2)$$

(since g is holomorphic when $\text{Im} z \in [-\varepsilon, \varepsilon]$.)

Note that

$$\int_{\gamma_2} f dz - \int_{\gamma_4} f dz = \sum_{n < R} \int_{C_{\varepsilon}^n} f dz = \sum_{n < R} 2\pi i \text{Res}(f, n) = -\sum_{n < R} \frac{n^5}{e^{2\pi n} - 1}. \quad (3)$$

Combining (1), (2), (3), we obtain

$$\sum_{n=1}^{\infty} \frac{n^5}{e^{2\pi n} - 1} = \int_0^{\infty} \frac{x^5}{e^{2\pi x} - 1} dx = \frac{1}{504}.$$

Another Solution

For exactly the same reasons, we have

$$\sum_{n=1}^{\infty} \frac{n^{4k+1}}{e^{2\pi n} - 1} = \int_0^{\infty} \frac{x^{4k+1}}{e^{2\pi x} - 1} dx = \frac{1}{2} \frac{B_{4k+2}}{4k+2} \forall k \geq 1.$$

This can be proved easily using the Eisenstein series.

Recall the Eisenstein series

$$E_k(\tau) = \sum_{\omega \in \Lambda^*} \omega^{-k} = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}$$

which is modular. Hence for $k = 4m + 2 \geq 6$, $E_k(i) = i^k E_k(i) = -E_k(i)$ so $E_k(i) = 0$, which implies

$$0 = 2\zeta(4m+2) - \frac{2(2\pi)^{4m+2}}{(4m+1)!} \sum_{r=1}^{\infty} \sigma_{4m+1}(r) e^{-2\pi r}.$$

Note that $\sum_{n=1}^{\infty} \frac{n^a q^n}{1-q^n} = \sum_{n=1}^{\infty} \sigma_a(n) q^n$, hence

$$\sum_{n=1}^{\infty} \frac{n^{4m+1}}{e^{2\pi n} - 1} = \sum_{r=1}^{\infty} \sigma_{4m+1}(r) e^{-2\pi r} = \frac{\zeta(4m+2)(4m+1)!}{2(2\pi)^{4m+2}} = \frac{1}{2} \frac{B_{4m+2}}{4m+2}.$$

Generalizations

It is well known that $E_4(i) = \frac{\Gamma(1/4)^8}{960\pi^4}$, so similar arguments show that

$$\sum_{n=1}^{\infty} \frac{n^3}{e^{2\pi n} - 1} = \frac{\Gamma(1/4)^8}{5120\pi^6} - \frac{1}{240},$$

and generally

$$\sum_{n=1}^{\infty} \frac{n^{4k-1}}{e^{2\pi n} - 1} = \frac{B_{4k}}{8k} (1 - \tilde{E}_{4k}(i)).$$

Since the modular group $M(\mathrm{SL}_2(\mathbb{Z}))$ is generated by E_4 and E_6 , and $E_6(i) = 0$, $\tilde{E}_{4k}(i) = q \cdot \tilde{E}_4(i)^k$ for some $q \in \mathbb{Q}$.